

# Soumen Ghosh, PhD

Medical Imaging AI & Quantitative Imaging

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## Research profile

Medical imaging AI researcher combining machine learning, structural and diffusion MRI, and quantitative CT. Research focuses on neuroimaging and neurological disease, occupational lung disease and thoracic imaging, and the evaluation of AI in clinical cohorts.

## Current appointments

**Senior Researcher, I-MED Radiology Network** | March 2026 - present

Clinical AI evaluation, quantitative imaging and occupational lung disease research.

**Honorary Fellow, The University of Queensland** | May 2025 - present

Neuroimaging and medical imaging AI research.

**Adjunct Lecturer, Charles Darwin University** | September 2026 - present

## Previous research appointments

**Research Officer, QIMR Berghofer Medical Research Institute** | July 2024 - February 2026

**Postdoctoral Research Fellow, The University of Queensland** | October 2023 - June 2024

**Visiting Researcher, Royal Brisbane and Women's Hospital** | February 2020 - September 2023

**Junior Research Fellow, IIIT Sri City** | August 2017 - July 2019

**Project Assistant, University of Hyderabad** | December 2016 - June 2017

## Education

**PhD, The University of Queensland** | 2024

Thesis: *Machine learning and deep learning-based approaches in epilepsy.*

Supervisors: David Reutens and Viktor Vegh.

**M.Tech, Information Technology**, University of Hyderabad | 2016

**MCA**, West Bengal University of Technology | 2013

**BCA**, West Bengal University of Technology | 2010

## Teaching & supervision

Teaching experience at Kaplan Business School, James Cook University Brisbane, The University of Queensland, IIIT Sri City and the University of Hyderabad. Topics include statistics, data visualisation, data science, digital health and research methods.

Supervision of postgraduate and undergraduate research in medical imaging AI, including brain MRI lesion segmentation and reproducible image analysis workflows.

# Selected research outputs

Soumen Ghosh | Selected academic CV | September 2026

## Published journal articles and conference papers

**Ghosh S**, Jestin Hannan C, Vashistha R, Kundu P, Brosda S, Aoude LG, Lonie J, Nathanson A, Ng J, Barbour AP, Vegh V. Stress-testing cross-cancer generalizability of 3D nnU-Net for PET-CT tumour segmentation: multi-cohort evaluation with novel oesophageal and lung cancer datasets. *International Conference on Computer Vision and Image Processing (CVIP)*, 2025. Published conference paper. [Preprint](#).

Shokraei Fard A, Reutens DC, Ramsay SC, Goodman SJ, **Ghosh S**, Vegh V. Image synthesis of interictal SPECT from MRI and PET using machine learning. 2024. *Frontiers in Neurology* 15:1383773. [doi:10.3389/fneur.2024.1383773](#).

Moradi H, Vashistha R, **Ghosh S**, O'Brien K, Hammond A, Rominger A, Sari H, Shi K, Vegh V, Reutens DC. Automated extraction of the arterial input function from brain images for parametric PET studies. 2024. *EJNMMI Research* 14:33. [doi:10.1186/s13550-024-01100-x](#).

Basha SHS, **Ghosh S**, Babu K, Dubey SR, Pulabaigari V, Mukherjee S. RCCNet: An Efficient Convolutional Neural Network for Histological Routine Colon Cancer Nuclei Classification. *ICARCV*, 2018, pp. 1222-1227. [Preprint](#).

## Accepted work and preprints

**Ghosh S**, Arya AS, Goswami T, Mandal S, Phamnguyen J, Vashistha R. CRIL-U-Net: Compact Ratio-Interaction Learning for Focal Cortical Dysplasia Segmentation from T1w and FLAIR MRI. Accepted at *MICAD 2026*. [arXiv:2608.03185](#).

**Ghosh S**, Phamnguyen J, Arya AS, Mandal S, Goswami T, Vashistha R. Benchmarking MRI Representations for Deep Learning-Based Focal Cortical Dysplasia Segmentation. 2026 preprint. [arXiv:2607.15605](#).

## Selected recognition

UQ travel support to attend OHBM (2024) and AES (2023).

MICCAI Student Participation Award (2021).

PReMI-Springer Student Award (2017).

## Research methods

Structural and diffusion MRI; quantitative CT; image segmentation; deep learning; clinical data curation and imaging quality control; multi-site harmonisation; diagnostic performance evaluation; reproducible analysis pipelines.

A fuller publication and conference abstract list is available on the [academic website](#).